Jonathan Liu

Seattle, WA | jliu1401@uw.edu | +1 (206) 589-0963
linkedin.com/in/jonathan-liu-202081252 | github.com/JonathanLiu1401 | jonathanliu1401.github.io

EDUCATION

University of Washington **Seattle, WA**
Bachelor of Science in Electrical and Computer Engineering | GPA: 3.93 *Sep 2024 – June 2027 (expected)*
Master of Science in Electrical and Computer Engineering (Combined BS/MS, Analog and Mixed-Signal) *Sep 2027 – June 2028*

- **Honors:** Frey Endowed EE Scholarship (2026–27), IEEE HKN Honor Society (Top 8% ECE), Tau Beta Pi Engineering Honor Society (Top 12.5%), Annual Dean's List (2024–2026).
- **Relevant Coursework: Analog IC Design (EE 435)**, **VLSI I (EE 476)**, Computer Architecture (EE 469), all in progress.

ANALOG IC DESIGN

Analog IC Tapeout, UW Analog Tapeout Sequence (EE 435/436/437) *Sep 2026 – Summer 2027*

- Designing an **analog IC for industrial-foundry fabrication** in a 3-quarter sequence sponsored by **Apple, AMD, and TSMC**.

Amplifier and Optical-Receiver Design in 45nm CMOS (EE 332) *Spring 2026*

- Designed an **operational transconductance amplifier (OTA)** in **Cadence Virtuoso** with **gm/I_D** sizing, then reused it as a **transimpedance amplifier** for an **optical receiver** with **6x** the bandwidth of a resistor-only front end.

Semiconductor Engineering Practicum **July – Aug 2026**
Hiroshima University *Hiroshima, Japan*

- Selected for a U.S.–Japan semiconductor consortium with **Micron Technology** and **Tokyo Electron**; hand-drew a **CMOS inverter layout** in **Siemens L-Edit**, fabricated it in a cleanroom, and confirmed working transistors on a **Keysight B1500A**.

RELEVANT EXPERIENCE

Analog Design Research Assistant **Sep 2026 – Present**
EMiT Lab, University of Washington *Seattle, WA*

- Built an **AI agent workflow for analog design in Cadence Virtuoso** that takes circuits from schematic through DRC, LVS, and parasitic extraction, plus a benchmark of four AI design tools; **the lab adopted the tool it recommended**.

Hardware Engineering Intern **June – Sep 2026**
Trellis Robotics *Mountain View, CA*

- **Designed a BMS and motor driver board from scratch** for a pipe-inspection robot after diagnosing why the previous design destroyed itself at power-up; the new board worked **the day it arrived with no hardware rework**.
- Engineered **battery protection and precision current-sensing circuits** for the robot's lithium battery, then fixed shutdown and power-switch behavior by reprogramming a battery-monitor chip's built-in settings, **avoiding a costly board respin**.
- Architected a **Power-over-Ethernet robot board** that powers two motors and a motion sensor over one network cable, and caught a chip timing flaw that corrupted network data **before fabrication**.
- Developed a **novel 3D positioning system** that maps a robot's path inside metal pipe, **where GPS and compasses fail**, using only a free-floating motion sensor and tether length; achieved **centimeter-level accuracy**.
- Built a **company-wide AI engineering assistant** that answers from the robot's code and docs via **retrieval-augmented generation (RAG)**, with a second AI model fact-checking every answer, **onboarding the entire company**.

Hardware Lead **Oct 2024 – Present**
Husky Robotics Drone Subteam *Seattle, WA*

- Led a 20+ person EE team and managed three custom PCBs in parallel: a **flight controller**, an electronic speed controller, and a power distribution board, including design reviews and the power and signal interfaces between them.
- Designed and **fabricated** a 4-layer **STM32H743** flight controller in **Altium**: schematic, layout, 5 V/3.3 V regulation, current sensing, and the connector pinout for motors, radios, and sensors.
- Led **electrical system integration** for a University Rover Challenge quadrotor: battery power, motor controllers, Pixhawk 6C flight computer, Jetson Orin Nano, and **MAVLink**.

PROJECTS

- **AC-DC Boost Converter (June 2026):** designed, simulated in **LTspice**, and **breadboarded** a two-stage converter delivering an adjustable **+10–21 VDC** rail at **0.4%** load regulation under **100 mV** ripple.
- **FPGARhythm (March 2026):** built a **SystemVerilog** rhythm game on an Intel **Cyclone V** FPGA.

SKILLS

IC Design & Test: Cadence Virtuoso, Spectre, ADE Assembler, Calibre DRC/LVS, Quantus PEX, gm/I_D sizing, wafer probing.
PCB & Embedded: Altium, KiCad, multi-rail PDN, STM32, ESP-IDF, I^2C/SPI/CAN, BMS, motor drivers, DC-DC converters.
Digital & Software: SystemVerilog, FPGA, C/C++, Python, SKILL, MATLAB, Linux, Git, LLM agents, RAG pipelines.
Leadership: President, IEEE UW Student Chapter; **Events Officer**, Hong Kong Student Association at UW.
Languages: English, Mandarin, Cantonese, Japanese.